

Machine Learning Fundamentals

1. Introduction to Machine Learning

Machine learning (ML) is a branch of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. There are three main types: supervised learning, unsupervised learning, and reinforcement learning.

2. Supervised Learning

In supervised learning, the algorithm is trained on labeled data. Common algorithms include: Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Neural Networks. Applications include spam detection, image classification, and credit scoring.

Machine Learning Fundamentals

3. Model Evaluation Metrics

Key metrics: Accuracy (fraction of correct predictions), Precision ($TP/(TP+FP)$), Recall ($TP/(TP+FN)$), F1-Score (harmonic mean of precision and recall), AUC-ROC (area under ROC curve), Mean Squared Error for regression tasks.

4. Overfitting and Regularisation

Overfitting occurs when a model learns noise in training data and fails to generalise. Regularisation techniques: L1 Lasso (drives weights to zero), L2 Ridge (shrinks weights), Dropout (deactivates neurons randomly), Early Stopping, Data Augmentation.

Machine Learning Fundamentals

5. Requirements for a Production ML System

A production ML system must satisfy: (1) Data Quality - clean and unbiased training data, (2) Reproducibility - experiments must be reproducible, (3) Scalability - handle growing data, (4) Monitoring - track model performance drift, (5) Retraining Pipeline - automated retraining, (6) Explainability - interpretable decisions, (7) Security - protection against adversarial attacks, (8) Compliance - GDPR, HIPAA and other regulations must be respected.